HSC Mathematics Extension 1 & 2: Probability Distributions

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Download PDF and resources:
`https://github.com/vuhung16au/math-olympiad-ml`

The theory of probabilities is at bottom nothing but common sense reduced to calculus; it enables us to appreciate with exactness that which accurate minds feel with a sort of instinct.

Pierre-Simon Laplace

Contents

1 Introduction

1.1 Project Overview

This booklet is a growing collection of probability distribution problems for NSW HSC Mathematics Extension 2 students. It covers continuous probability distributions (including the normal distribution), binomial distributions, and selected advanced topics such as Monte Carlo simulations and the hypergeometric distribution.

1.2 Target Audience

This resource is written for students who want extra practice with probability distributions in a clear, classroom-friendly format. Every problem begins with an upside-down **Hints** panel and ends with full worked solutions followed by short takeaways.

1.3 How to Use This Booklet

- Read the introductory theory for each topic before attempting problems.
- Attempt each problem before checking the hint box, then read the solution.
- Pay attention to whether a question involves a discrete or continuous distribution.
- Use the takeaways after each solution to reinforce key ideas.

1.4 Extension 1 Knowledge Needed

To work through this booklet, students should be comfortable with:

- **Basic probability:** sample spaces, complementary events, conditional probability.
- **Algebra:** solving equations, working with exponentials and logarithms.
- **Calculus:** integration, including integration by substitution.
- **Binomial theorem:** expansion of $(1+x)^n$ and binomial coefficients $\binom{n}{r}$.
- **Statistics basics:** mean, variance, standard deviation.

1.5 Random experiments and random variables

Throwing four fair coins and recording the number of heads is a **random experiment** because more than one outcome is possible. Throwing four coins into an empty bucket and counting how many coins are in the bucket is a **deterministic experiment** because the outcome is necessarily 4.

A **(discrete) random variable** attaches a numerical value to each outcome of such an experiment.

1.6 Discrete probability distributions

Suppose the sample space of an experiment can be listed (written down in order) and the possible outcomes are numeric. Those outcomes together with their probabilities form a **discrete probability distribution**.

Probability mass function (PMF). If X takes values $x_1, x_2, \ldots$ then the PMF is $p(x) = P(X = x)$ where $p(x_i) \geq 0$ and $\sum_i p(x_i) = 1$. For example, let X be the number of heads in three fair tosses of a coin. Then $X \in \{0, 1, 2, 3\}$ with

$$P(X = 0) = \tfrac{1}{8}, \qquad P(X = 1) = \tfrac{3}{8}, \qquad P(X = 2) = \tfrac{3}{8}, \qquad P(X = 3) = \tfrac{1}{8}.$$

Each probability is computed by counting equiprobable sequences (e.g. $P(X = 2) = \binom{3}{2}(\frac{1}{2})^3$).

1.7 Uniform discrete distributions

A discrete distribution is **uniform** when every outcome in its sample space has the same probability — the values are equally likely. For instance, rolling a fair six-sided die and recording the upper face produces a uniform distribution on $\{1, 2, 3, 4, 5, 6\}$ with $P(X = k) = \frac{1}{6}$ for each k.

1.8 Bernoulli trials and binomial experiments

A **Bernoulli trial** is a single random experiment with exactly two outcomes, often called "success" and "failure", with probabilities p and $q = 1 - p$. Tossing a coin once with success = heads gives $p = \frac{1}{2}$. Throwing a die once and recording only whether a six appears is a Bernoulli trial with $p = \frac{1}{6}$ if success is "six".

A **binomial experiment** consists of n independent Bernoulli trials, each with the same success probability p, and the random variable X is the total number of successes (order of successes does not matter). For example, tossing the same coin five times and counting heads is a binomial experiment with $n = 5$. If each toss is independent with probability p of heads,

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

The table below lists outcomes for five fair tosses ($p = \frac{1}{2}$):

k (heads)	0	1	2	3	4	5
$P(X = k)$	$\frac{1}{32}$	$\frac{5}{32}$	$\frac{10}{32}$	$\frac{10}{32}$	$\frac{5}{32}$	$\frac{1}{32}$

1.9 Improper integrals

For continuous models, probabilities often use integrals on unbounded intervals, for example

$$P(X \leq a) = \int_{-\infty}^{a} f(x)\, dx, \qquad P(X \in \mathbb{R}) = \int_{-\infty}^{\infty} f(x)\, dx.$$

Such **improper integrals** are evaluated as limits of proper integrals (e.g. $\int_{-\infty}^{\infty} = \lim_{b\to\infty} \int_{-b}^{b}$ when the limit exists). Checking that $\int_{-\infty}^{\infty} f(x)\, dx = 1$ confirms that f can be a valid pdf.

1.10 A first look at the Central Limit Theorem

Let independent observations $X_1, \ldots, X_n$ share mean μ and variance σ^2, and define the sample mean $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Under mild conditions, for large n the distribution of $\overline{X}$ is approximately normal with

$$E(\overline{X}) = \mu, \qquad \mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}, \qquad \text{s.d.}(\overline{X}) = \frac{\sigma}{\sqrt{n}}.$$

Central limit idea: the distribution of means of many independent draws becomes approximately normal even when the original variable is not. That is why the normal distribution appears throughout statistics, for both discrete and continuous models.

1.11 Gaussian (normal) distributions and the empirical rule

The **normal** (Gaussian) family includes the standard normal $Z \sim N(0,1)$ and $X \sim N(\mu, \sigma^2)$ with pdf the familiar bell curve. The mean locates the centre; the standard deviation σ measures spread.

Empirical rule (68–95–99.7). For a normal distribution, approximately

- 68% of values lie within one standard deviation of the mean ($\mu \pm \sigma$);
- 95% within two standard deviations ($\mu \pm 2\sigma$);
- 99.7% within three standard deviations ($\mu \pm 3\sigma$).

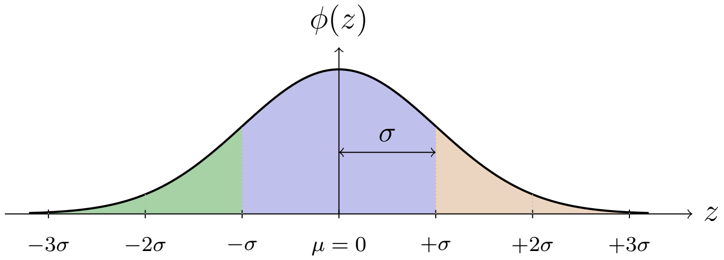

Figure: standard normal curve $z \mapsto \phi(z)$ with mean at 0, standard-deviation markers on the z-axis, and shaded bands illustrating central and tail regions.

1.12 Core Formulas and Ideas

Probability density function (pdf). A function $f(x)$ is a probability density function if

$$f(x) \geq 0 \quad \text{for all } x, \qquad \text{and} \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1.$$

The probability that X lies in the interval $[a, b]$ is

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx.$$

Mean (expected value). For a continuous random variable with pdf $f(x)$,

$$\mu = E(X) = \int_{-\infty}^{\infty} x\,f(x)\,dx.$$

Variance.

$$\sigma^2 = \mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2.$$

Standard deviation. $\sigma = \sqrt{\mathrm{Var}(X)}$.

Standard normal distribution. The standard normal distribution $Z \sim N(0,1)$ has pdf

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

General normal distribution. If $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0,1).$$

Binomial distribution. If $X \sim B(n,p)$, then

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}, \quad r = 0, 1, \ldots, n.$$

The mean is $\mu = np$ and the variance is $\sigma^2 = np(1-p)$.

Normal approximation to binomial. For large n,

$$X \sim B(n,p) \approx N(np,\, np(1-p)) .$$

Sample proportion. For a sample of size n from a population with proportion p,

$$\hat{p} = \frac{X}{n}, \qquad E(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

2 Continuous Probability Distributions

2.1 Relative Frequency

Relative frequency is the proportion of times an event occurs in a repeated experiment. If an event occurs k times in n trials, its relative frequency is

$$\text{relative frequency} = \frac{k}{n}.$$

As the number of trials n increases, the relative frequency of an event tends towards its theoretical probability. This is the basis of the frequency interpretation of probability.

Remark 2.1

Histograms of relative frequencies approximate continuous probability density functions as the class width decreases and the sample size increases.

2.2 Continuous Distributions

A **continuous random variable** X can take any value in an interval (or union of intervals) of real numbers. Instead of assigning probability to individual values, we use a **probability density function** (pdf) $f(x)$.

Properties of a pdf:

- $f(x) \geq 0$ for all x.
- $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = 1$.

- $P(a \le X \le b) = \int_a^b f(x)\,dx$.

Note that for a continuous random variable, $P(X = c) = 0$ for any particular value c. This means $P(a \le X \le b) = P(a < X < b)$.

Cumulative distribution function (CDF). The CDF is defined by

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt.$$

The pdf is the derivative of the CDF: $f(x) = F'(x)$.

2.3 Mean and Variance of a Distribution

Mean (Expected Value). The mean of a continuous random variable X with pdf $f(x)$ is

$$\mu = E(X) = \int_{-\infty}^{\infty} x\,f(x)\,dx.$$

Variance. The variance measures the spread of the distribution:

$$\sigma^2 = \mathrm{Var}(X) = E\big[(X-\mu)^2\big] = \int_{-\infty}^{\infty} (x-\mu)^2\,f(x)\,dx.$$

An equivalent and often more convenient formula is

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2.$$

Standard deviation. $\sigma = \sqrt{\mathrm{Var}(X)}$.

Properties of expectation:

- $E(aX + b) = aE(X) + b$ for constants a, b.
- $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$.

2.4 The Standard Normal Distribution

The **standard normal distribution** is denoted $Z \sim N(0, 1)$. It has mean $\mu = 0$ and standard deviation $\sigma = 1$.

Its pdf is

$$\phi(z) = \frac{1}{\sqrt{2\pi}}\,e^{-z^2/2}, \qquad -\infty < z < \infty.$$

The cumulative distribution function is

$$\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \phi(t)\,dt.$$

Values of $\Phi(z)$ are read from the standard normal table.

Symmetry property: $P(Z \le -z) = 1 - P(Z \le z)$, or equivalently $\Phi(-z) = 1 - \Phi(z)$.

Key values:

- $P(-1 \le Z \le 1) \approx 0.6827$
- $P(-2 \le Z \le 2) \approx 0.9545$
- $P(-3 \le Z \le 3) \approx 0.9973$

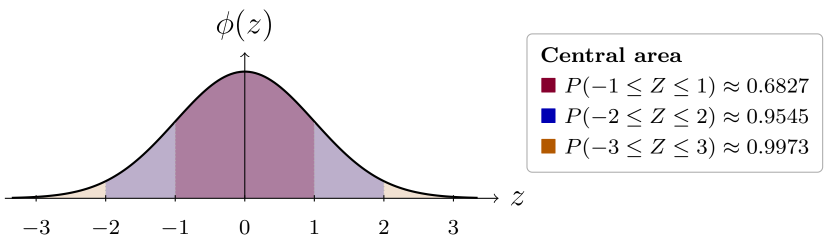

Figure: the standard normal curve with the central one-, two-, and three-standard-deviation regions shaded.

2.5 General Normal Distributions

If $X \sim N(\mu, \sigma^2)$, the pdf of X is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Standardising. To compute probabilities, convert X to the standard normal variable Z:

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1).$$

Then

$$P(a \le X \le b) = P\left(\frac{a-\mu}{\sigma} \le Z \le \frac{b-\mu}{\sigma}\right).$$

The 68-95-99.7 rule. For any normal distribution:

- approximately 68% of values lie within one standard deviation of the mean.
- approximately 95% of values lie within two standard deviations of the mean.
- approximately 99.7% of values lie within three standard deviations of the mean.

2.6 Applications of the Normal Distribution

The normal distribution models many naturally occurring quantities, including:

- heights and weights in a population,
- measurement errors in scientific experiments,
- scores on standardised tests,
- lifetimes of manufactured components.

Finding probabilities. To find $P(X \le k)$ for $X \sim N(\mu, \sigma^2)$:

1. Standardise: compute $z = \dfrac{k-\mu}{\sigma}$.
2. Read $\Phi(z)$ from the standard normal table.

Finding cut-off values. Given $P(X \le k) = p$, find k:

1. Find z such that $\Phi(z) = p$ (using the inverse normal table or technology).
2. Convert back: $k = \mu + z\sigma$.

2.7 Investigations Using the Normal Distribution

This section develops investigative skills with the normal distribution, including:

- estimating population parameters from sample data,
- testing whether a data set is approximately normally distributed (e.g. using a histogram or normal probability plot),
- interpreting results in context.

Checking normality. A data set may be approximately normal if:

- its histogram is roughly bell-shaped and symmetric,
- approximately 68%, 95%, and 99.7% of data lie within 1, 2, and 3 standard deviations of the mean respectively.

3 Binomial Distributions

3.1 Binomial Probability

A **binomial experiment** has the following properties:

- A fixed number n of trials.
- Each trial has exactly two outcomes: *success* or *failure*.
- The probability of success p is the same on every trial.
- The trials are independent.

If X is the number of successes in n independent trials with probability p of success, then

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}, \qquad r = 0, 1, 2, \ldots, n.$$

Cumulative binomial probability:

$$P(X \le k) = \sum_{r=0}^{k} \binom{n}{r} p^r (1-p)^{n-r}.$$

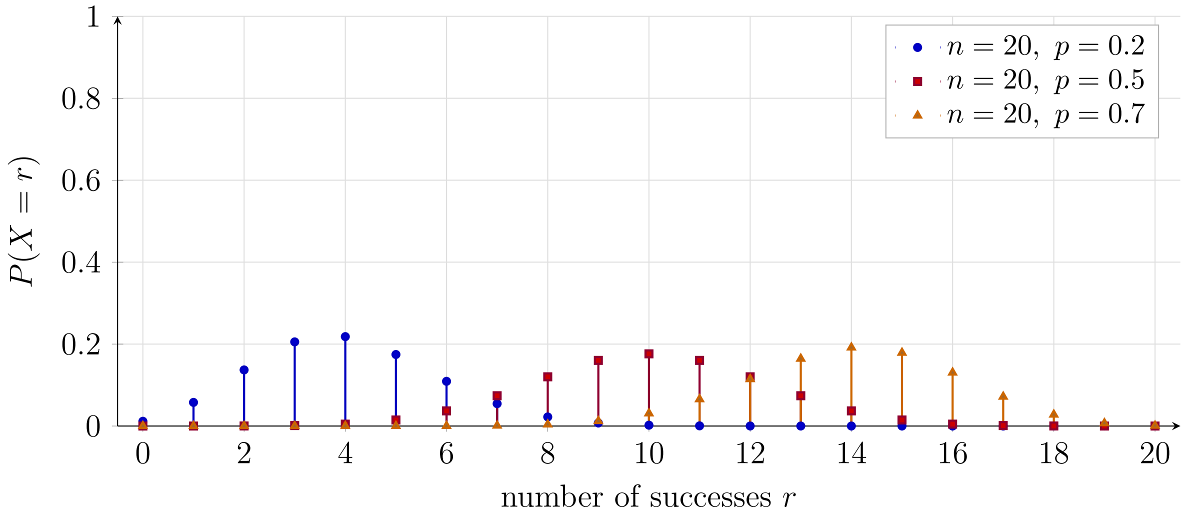

Figure: binomial probability mass functions for $X \sim B(20, p)$. As p increases, the distribution shifts to the right because the expected number of successes is np.

3.2 Binomial Distributions

If $X \sim B(n, p)$, we say X follows a **binomial distribution** with parameters n (number of trials) and p (probability of success).

Mean and variance of the binomial distribution:

$$E(X) = np, \qquad \mathrm{Var}(X) = np(1-p).$$

Shape of the distribution:

- When $p = 0.5$, the distribution is symmetric.
- When $p < 0.5$, the distribution is right-skewed (positively skewed).
- When $p > 0.5$, the distribution is left-skewed (negatively skewed).
- As n increases, the distribution becomes more bell-shaped.

Remark 3.1

The binomial distribution is the most important discrete distribution in the HSC syllabus. Make sure you can correctly identify n, p, and r before substituting into the formula.

3.3 Normal Approximations to a Binomial

When n is large, the binomial distribution $B(n, p)$ can be approximated by a normal distribution. Specifically,

$$X \sim B(n, p) \approx N\left(\mu, \sigma^2\right), \qquad \mu = np, \quad \sigma^2 = np(1-p).$$

When is the approximation valid? A common guideline is that both $np \geq 5$ and $n(1-p) \geq 5$.

Continuity correction. Because the binomial is discrete and the normal is continuous, a continuity correction improves accuracy:

$$P(X = k) \approx P\left(k - \tfrac{1}{2} \leq X \leq k + \tfrac{1}{2}\right).$$

For example:

$$P(X \leq k) \approx P\left(X \leq k + \tfrac{1}{2}\right) \quad \text{(normal approximation)}.$$

3.4 Sample Proportions

Given a random sample of size n from a population with true proportion p, the **sample proportion** is

$$\hat{p} = \frac{X}{n},$$

where $X \sim B(n, p)$ is the number of successes in the sample.

Mean and variance of $\hat{p}$:

$$E(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

Approximate distribution. For large n,

$$\hat{p} \approx N\left(p, \ \frac{p(1-p)}{n}\right).$$

Confidence intervals. An approximate 95% confidence interval for p is

$$\hat{p} \pm 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

4 Advanced Topics

(These topics are outside the HSC Mathematics Extension 2 syllabus but provide valuable enrichment.)

4.1 Monte Carlo Simulations

Monte Carlo simulation is a computational technique that uses random sampling to approximate numerical results. It is particularly useful when a problem is too complex to solve analytically.

Basic idea:

1. Define a random experiment whose outcome is related to the quantity you want to estimate.
2. Simulate the experiment a large number of times (e.g. $N = 10{,}000$ trials).
3. Estimate the quantity as the proportion (or average) of the simulated outcomes.

Estimating π by Monte Carlo. Consider a unit square with an inscribed quarter-circle of radius 1.

- Generate N random points (x_i, y_i) with $x_i, y_i \in [0, 1]$.
- Count the number M of points satisfying $x_i^2 + y_i^2 \leq 1$.
- Then $\pi \approx 4M/N$.

Why it works. The probability that a random point in the unit square lies inside the quarter-circle is $\pi/4$, since the area of the quarter-circle is $\pi/4$ and the area of the square is 1.

4.2 Hypergeometric Distribution

The **hypergeometric distribution** models the number of successes in a sample drawn *without replacement* from a finite population.

Setup. A population of N items contains K successes (and $N - K$ failures). A sample of n items is drawn without replacement. Let X be the number of successes in the sample.

Probability formula:

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \qquad \max(0,\, n - (N - K)) \leq k \leq \min(n, K).$$

Mean and variance:

$$E(X) = \frac{nK}{N}, \qquad \mathrm{Var}(X) = \frac{nK(N-K)(N-n)}{N^2(N-1)}.$$

Contrast with binomial. The binomial distribution models sampling *with replacement*; the hypergeometric distribution models sampling *without replacement*. For large N, the two distributions are approximately the same.

Remark 4.1

The hypergeometric distribution arises naturally in quality control, card problems, and ecological sampling.

4.3 Simulating the Central Limit Theorem (spreadsheet lab)

This activity is enrichment and is not part of the HSC Mathematics Extension 2 examination. It illustrates the Central Limit Theorem by simulation in Google Sheets.

In this challenge you use Google Sheets to build a highly skewed "population", draw repeated random samples, and see that the *distribution of sample means* becomes approximately normal.

Generating the population. In a new sheet, label `A1 Population`. In cell `A2` enter
`=INT(-LN(RAND())*20)`
Fill down to row 1001. This mimics nonnegative "waiting times" (roughly exponential and strongly right-skewed).

Tasks.

a. In `C1:C3`, report the population mean (`AVERAGE`), population standard deviation (`STDEV.P`), and skewness (`SKEW`) of column `A`. Skew above about 1 indicates a heavy right tail.

b. Draw samples of size $n = 30$ with replacement from the thousand rows. In `E2` enter the following (one line in the sheet):

```
=AVERAGE(ARRAYFORMULA(INDEX($A$2:$A$1001, RANDARRAY(30, 1, 1, 1000, TRUE))))
```

Fill from `E2` down to `E201` to obtain 200 sample means.

c. In `G1:G2`, compute the mean and sample standard deviation of the 200 means in column `E`. Compare to the CLT predictions $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \sigma/\sqrt{n}$ with $n = 30$ (use μ, σ from part (a)).

d. List bin edges in column `I` (e.g. $0, 1, \dots, 20$). In `J2` enter `FREQUENCY` for the range `E2:E201` with those bins, then build a column histogram. The means should follow a bell-shaped curve even though the raw data in `A` are skewed.

What to expect. The mean of the 200 sample means should match the population mean closely; the spread of those means should match $\sigma/\sqrt{30}$. The histogram of means is much more symmetric than the original population — typical CLT behaviour for moderate n.

Notes for teachers. `RANDARRAY` returns random indices; `INDEX` pulls population values; `AVERAGE` forms each mean. `FREQUENCY` is an array function in Sheets; enter it once and let results spill, or use your spreadsheet's equivalent.

5 Part 1: Worked Problems

Part 1 contains three sets of problems with detailed solutions. Each solution includes a strategy note, step-by-step working, and a takeaways box.

5.1 Basic

Problem 5.1: Observing random number generation

A student uses a computer's random number generator to output 30 independent random integers from the set $\{0, 1, 2, \ldots, 9\}$. The generator is designed to be fair, so each digit should be equally likely in the long run, but the student only observes a short sample.

(i) State the theoretical probability distribution for a single generated digit. What is the expected frequency for each digit in a sample of size 30?

(ii) The observed frequencies for digits 0 through 9 are, in order, `[2, 4, 3, 1, 5, 2, 4, 3, 2, 4]`. Calculate the empirical probability of an even digit (treating 0 as even) and compare to the theoretical probability of an even digit.

(iii) Name the principle that explains why empirical frequencies need not match expectations in a small sample. What happens if 30,000 digits are generated instead?

Hint:

- Part (i): Under a fair uniform model, multiply the number of trials by the probability of one outcome to get expected counts.
- Part (ii): Sum frequencies for even digits $0, 2, 4, 6, 8$; divide by 30. The theoretical chance of an even digit is $5/10$.
- Part (iii): Compare short-run fluctuation with long-run stability (Law of Large Numbers).

Solution 5.1

(i) Let X be one generated digit. Under the fair model,

$$P(X = x) = \frac{1}{10} = 0.1, \qquad x = 0, 1, \ldots, 9.$$

In 30 independent generated digits, the expected frequency of any particular digit is

$$30 \times 0.1 = 3.$$

This does not mean every digit must appear exactly three times in a short run; it is the long-run average count.

(ii) The even digits are $0, 2, 4, 6, 8$. Their observed frequencies are

$$2 + 3 + 5 + 4 + 2 = 16.$$

Hence the empirical probability of an even digit is

$$\frac{16}{30} \approx 0.533.$$

The theoretical probability is $\frac{5}{10} = 0.5$, so this particular sample has a slightly higher even-digit proportion than expected.

(iii) The mismatch is explained by sampling variability. The **Law of Large Numbers** says that as the number of generated digits becomes very large, the relative frequencies should settle closer to the theoretical probabilities. With 30,000 draws, each digit's relative frequency should be much closer to 0.1, and the even-digit relative frequency should be much closer to 0.5.

Takeaways 5.1

- Expected frequency is a long-run average, not a quota in every short run.
- Empirical probability reflects data; theoretical probability reflects the model.
- Large samples smooth out random variation.

Problem 5.2: Linear transformations and variance

Let X be a random variable with mean μ and variance σ^2, where

$$\mathrm{Var}(X) = E[(X-\mu)^2].$$

This problem proves three formulas that are used constantly in probability: the computational variance formula, the effect of a linear transformation, and the standardisation rule. You may use linearity of expectation: $E(aX) = aE(X)$ and $E(X+b) = E(X)+b$.

(i) Prove $\mathrm{Var}(X) = E(X^2) - \mu^2$.

(ii) For $Y = aX + b$ with constants a, b, prove $\mathrm{Var}(Y) = a^2\mathrm{Var}(X)$.

(iii) For $Z = \dfrac{X-\mu}{\sigma}$, prove $E(Z) = 0$ and $\mathrm{Var}(Z) = 1$.

Hint:

- Expand $(X-\mu)^2$ and apply $E[\cdot]$ term by term; treat μ as a constant.
- Write $\mathrm{Var}(Y) = E[(Y-E(Y))^2]$ with $E(Y) = a\mu + b$ and simplify $Y - E(Y) = a(X-\mu)$.
- View Z as $\frac{1}{\sigma}X - \frac{\mu}{\sigma}$ and use (ii).

Solution 5.2

(i) Start from the definition and expand the square:

$$\mathrm{Var}(X) = E[(X-\mu)^2] = E[X^2 - 2\mu X + \mu^2].$$

Using linearity of expectation and treating μ as a constant,

$$\mathrm{Var}(X) = E(X^2) - 2\mu E(X) + \mu^2.$$

Since $E(X) = \mu$, this becomes

$$\mathrm{Var}(X) = E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2.$$

(ii) If $Y = aX + b$, then $E(Y) = a\mu + b$. Therefore

$$Y - E(Y) = aX + b - (a\mu + b) = a(X-\mu).$$

So

$$\mathrm{Var}(Y) = E([a(X-\mu)]^2) = a^2E[(X-\mu)^2] = a^2\mathrm{Var}(X).$$

The shift b disappears because it moves every value and the mean by the same amount.
(iii) Here $a = 1/\sigma$, $b = -\mu/\sigma$. Then $E(Z) = \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0$, and $\mathrm{Var}(Z) = (1/\sigma)^2\sigma^2 = 1$.

Takeaways 5.2

- $E[\cdot]$ is linear: sums and constants pass through cleanly.
- Adding b shifts the mean only; multiplying by a scales variance by a^2.
- Standardization always yields mean 0, variance 1.

Problem 5.3: The stepped probability density function

A continuous random variable X has a probability density function made from two horizontal steps:

$$f(x) = \begin{cases} k & 0 \le x < 3 \\ 2k & 3 \le x \le 6 \\ 0 & \text{otherwise.} \end{cases}$$

(i) Find k so that f is a valid pdf.

(ii) Calculate $E(X)$.

(iii) Calculate $\mathrm{Var}(X)$.

Hint:

- Integrate over $[0, 3]$ and $[3, 6]$ and equate total area to 1.
- Use $\int x f(x)\, dx$ split over the same pieces.
- $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$.

Solution 5.3

(i) A valid pdf must have total area 1. Here the area is the sum of two rectangles:

$$\int_0^3 k\,dx + \int_3^6 2k\,dx = 3k + 6k = 9k.$$

Thus $9k = 1$, so $k = \frac{1}{9}$.
(ii)

$$E(X) = \int_0^3 x\left(\frac{1}{9}\right)\,dx + \int_3^6 x\left(\frac{2}{9}\right)\,dx.$$

Evaluating,

$$E(X) = \left[\frac{x^2}{18}\right]_0^3 + \left[\frac{x^2}{9}\right]_3^6 = \frac{9}{18} + \left(\frac{36}{9} - \frac{9}{9}\right) = 3.5.$$

The mean is pulled to the right of the midpoint 3 because the density is twice as high on $[3, 6]$.
(iii) First compute

$$E(X^2) = \int_0^3 x^2\left(\frac{1}{9}\right)\,dx + \int_3^6 x^2\left(\frac{2}{9}\right)\,dx.$$

This gives

$$E(X^2) = \left[\frac{x^3}{27}\right]_0^3 + \left[\frac{2x^3}{27}\right]_3^6 = 1 + 14 = 15.$$

Therefore

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 15 - (3.5)^2 = \boxed{2.75}.$$

Takeaways 5.3

- Rectangular strips make k solvable by areas of rectangles.
- Split integrals wherever the formula for f changes.
- Always use $E(X^2) - [E(X)]^2$ for variance after $E(X^2)$ is found.

5.2 Medium

Problem 5.4: The transformed uniform distribution

Let X be discrete uniform on the first n positive odd integers

$$1, 3, 5, \ldots, 2n-1.$$

For comparison, let U be discrete uniform on $\{1, 2, \ldots, n\}$, with

$$E(U) = \frac{n+1}{2}, \qquad \mathrm{Var}(U) = \frac{n^2-1}{12}.$$

The key idea is to recognise the odd integers as a linear transformation of the simpler variable U.

(i) Give $P(X = x)$ for outcomes in the sample space.

(ii) Write X as a linear function of U and find $E(X)$.

(iii) Find $\mathrm{Var}(X)$ in terms of n.

(iv) Let $S = X_1 + X_2$ be the sum of two independent draws. Find $E(S)$, $\mathrm{Var}(S)$, and $P(S \text{ is maximal})$.

Hint:

- Odd integers are $2k-1$: relate X to U by $X = 2U - 1$.
- Use $\mathrm{Var}(aU + b) = a^2\mathrm{Var}(U)$.
- Maximal S arises only when both draws equal $2n-1$.

Solution 5.4

(i) n equally likely values: $P(X=x)=\frac{1}{n}$ for each allowed x.
(ii) $X=2U-1$, so $E(X)=2E(U)-1=n$.
(iii) $\mathrm{Var}(X)=\mathrm{Var}(2U-1)=4\mathrm{Var}(U)=\frac{n^2-1}{3}$.
(iv) Since X_1 and X_2 are independent copies of X,

$$E(S)=E(X_1)+E(X_2)=n+n=2n.$$

Independence also lets us add variances:

$$\mathrm{Var}(S)=\mathrm{Var}(X_1)+\mathrm{Var}(X_2)=\frac{2(n^2-1)}{3}.$$

The largest possible sum is

$$(2n-1)+(2n-1)=4n-2,$$

and this occurs only when both draws equal $2n-1$. Each has probability $1/n$, so

$$P(S\text{ is maximal})=\frac{1}{n}\cdot\frac{1}{n}=\frac{1}{n^2}.$$

Takeaways 5.4

- Affine maps $U\mapsto 2U-1$ avoid summing long arithmetic lists.
- Variance scales with the square of the multiplier on U.
- Sums of i.i.d. discrete uniforms have a peaked distribution—extreme totals are rare ($1/n^2$ here).

Problem 5.5: The asymmetric triangular distribution

A continuous random variable X has a triangular-shaped pdf which rises linearly from 0 to a peak at $x=1$, then falls linearly to 0 at $x=3$:

$$f(x)=\begin{cases} cx & 0\le x\le 1\\ c\left(\frac{3-x}{2}\right) & 1<x\le 3\\ 0 & \text{otherwise.}\end{cases}$$

(i) Find c, sketch $y=f(x)$, and check $f(x)\ge 0$.

(ii) Show total area is 1 by geometry and by integration.

(iii) Find the cdf $F(x)$.

(iv) Find the lower quartile Q_1 and median Q_2.

Hint:

- Triangular area on $[0, 3]$ with peak at $x = 1$.
- Split $\int_0^1$ and $\int_1^3$.
- Compare target probabilities to $F(1) = \frac{1}{3}$.
- Quadratic on the second branch: keep roots in $(1, 3]$.

Solution 5.5

(i) Area $\frac{1}{2} \cdot 3 \cdot f(1) = \frac{1}{2} \cdot 3 \cdot c = 1 \Rightarrow c = \frac{2}{3}$. Then $f(x) = \frac{2}{3}x$ on $[0, 1]$ and $\frac{1}{3}(3 - x)$ on $(1, 3]$ — nonnegative on $[0, 3]$.
(ii) Geometrically, the graph is a triangle with base length 3 and height $c = \frac{2}{3}$, so its area is

$$\frac{1}{2} \cdot 3 \cdot \frac{2}{3} = 1.$$

By integration,

$$\int_0^1 \frac{2}{3}x\,dx + \int_1^3 \frac{1}{3}(3 - x)\,dx = \frac{1}{3} + \frac{2}{3} = 1.$$

Both methods confirm that f has total area 1.
(iii) $F(x) = 0$ for $x < 0$; for $0 \le x \le 1$, $F(x) = \frac{x^2}{3}$; for $1 < x \le 3$, $F(x) = \frac{1}{3} + \int_1^x \frac{1}{3}(3 - t)\,dt = x - \frac{x^2}{6} - \frac{1}{2}$; for $x > 3$, $F = 1$.
(iv) Q_1 solves $\frac{x^2}{3} = 0.25$, so $Q_1 = \frac{\sqrt{3}}{2}$. For Q_2, $x - \frac{x^2}{6} - \frac{1}{2} = \frac{1}{2}$ yields $Q_2 = 3 - \sqrt{3}$.

Takeaways 5.5

- Piecewise pdfs splice cdfs continuously at joins.
- Check $F(1)$ before selecting which cdf branch solves $F = q$.
- Respect domain constraints when rejecting quadratic roots.

Problem 5.6: The variance identity and mean squared error

Let X have pdf $f(x)$ and mean μ. This problem proves two useful identities: the computational formula for variance and the decomposition of mean squared error. The final parts apply the identities to a symmetric density.

(i) Prove $\displaystyle\int_{-\infty}^{\infty} (x - \mu)^2 f(x)\,dx = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$.

(ii) Show $\displaystyle\int_{-\infty}^{\infty} (x - c)^2 f(x)\,dx = \mathrm{Var}(X) + (\mu - c)^2$.

(iii) For $f(x) = \frac{3}{4}(1 - x^2)$ on $[-1, 1]$, compute $\mathrm{Var}(X)$.

(iv) Minimize $E[(X - c)^2]$ over constants c for this f.

Hint:

- Expand $(x-\mu)^2$ and integrate termwise.
- Write $(x-c) = (x-\mu) + (\mu - c)$ and expand the square—the cross-term integrates to 0.
- f is even on $[-1, 1]$, hence $\mu = 0$.
- The expression in (ii) is minimized when $(\mu - c)^2 = 0$.

Solution 5.6

(i) Expand the square inside the integral:

$$\int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx = \int (x^2 - 2\mu x + \mu^2) f(x)\,dx.$$

Using $\int f(x)\,dx = 1$ and $\int x f(x)\,dx = \mu$, this becomes

$$\int x^2 f(x)\,dx - 2\mu^2 + \mu^2 = \int x^2 f(x)\,dx - \mu^2.$$

(ii) Write

$$x - c = (x-\mu) + (\mu - c).$$

After squaring and integrating,

$$E[(X-c)^2] = E[(X-\mu)^2] + 2(\mu - c)E(X-\mu) + (\mu - c)^2.$$

The middle term is 0 because $E(X-\mu) = 0$, so

$$E[(X-c)^2] = \mathrm{Var}(X) + (\mu - c)^2.$$

(iii) Symmetry yields $\mu = 0$. Thus $\mathrm{Var}(X) = E(X^2) = \frac{3}{4}\int_{-1}^{1} x^2(1-x^2)\,dx = \frac{1}{5}$.
(iv) Unique minimizer $c = \mu = 0$, with minimum $\mathrm{Var}(X) = \frac{1}{5}$.

Takeaways 5.6

- Computational variance $E(X^2) - \mu^2$ comes straight from expanding squares.
- MSE decomposition shows the mean minimizes expected squared error.
- Even pdf on symmetric support $\Rightarrow \mu = 0$.

Problem 5.7: Evaluating clinical trial efficacy

A drug has a baseline helpful-response rate of 70% under the usual delivery method. We use $p = 0.7$ as the reference probability and compare trial results against the binomial model using normal approximations.

(i) Team A: $n = 50$, 45 responders. Assuming $X \sim B(50, 0.70)$ under the null/reference, compute the approximate z-score for observing at least 45 successes using the normal approximation with *continuity correction*.

(ii) Team B expects exactly 80% successes in n patients ($0.8n \in \mathbb{Z}$). Ignoring corrections, determine the smallest n whose uncorrected z-score (relative to baseline $p = 0.7$) strictly exceeds Team A's *uncorrected* z-score for 45/50.

Hint:

- For ≥ 45, use boundary 44.5 versus $N(np, npq)$.
- Compare $z_B(n) = (0.8n - 0.7n)/\sqrt{0.21n}$ with $z_A = (45-35)/\sqrt{10.5}$, then enforce divisibility constraints.

Solution 5.7

(i) $\mu = 35$, $\sigma^2 = 10.5$. With correction, $z = \frac{44.5-35}{\sqrt{10.5}} = \frac{9.5}{\sqrt{10.5}} \approx 2.93$.
(ii) For comparison, Team A's uncorrected score is

$$z_A = \frac{45-35}{\sqrt{10.5}} = \frac{10}{\sqrt{10.5}}.$$

If Team B has exactly $0.8n$ responders, its uncorrected score relative to the baseline mean $0.7n$ is

$$z_B(n) = \frac{0.8n - 0.7n}{\sqrt{0.7(0.3)n}} = \frac{0.1n}{\sqrt{0.21n}} = \sqrt{\frac{n}{21}}.$$

We require $z_B(n) > z_A$, so

$$\sqrt{\frac{n}{21}} > \frac{10}{\sqrt{10.5}} \implies n > 200.$$

Also $0.8n$ must be an integer, so n must be a multiple of 5. The smallest valid value greater than 200 is $\boxed{205}$.

Takeaways 5.7

- Continuity corrections matter when n is modest.
- Evidence strength depends both on proportion and effective sample size.
- Algebraic isolation of $\sqrt{n}$ simplifies comparisons cleanly.

Problem 5.8: The musical fundraiser

At a school musical fundraiser, attendees spent different amounts on tickets, snacks, and small donations. Instead of recording every exact amount, the organisers grouped the spending into \$10 intervals. In the table below, each **class** is a spending interval, the **centre** x_i is the midpoint used to represent everyone in that interval, and the **frequency** f_i is the number of attendees in that interval.
Grouped spending (\$):

class	$0 \le x < 10$	$10 \le x < 20$	$20 \le x < 30$	$30 \le x < 40$	$40 \le x < 50$
centre x_i	5	15	25	35	45
frequency f_i	30	a	45	b	15

There are $\sum f_i = 150$ attendees with mean \$21 estimated from class centres.

(i) Find a and b.

(ii) Find $\text{Var}(X)$ using class centres, and estimate the median amount spent using cumulative frequencies.

(iii) Assuming uniformity within classes, approximate $P(X > 28)$.

(iv) For profit $P = 0.6X - 5$, find $E(P)$ and $\sigma(P)$.

Hint:

- Use $\sum f = 150$ and $\frac{\sum fx}{\sum f} = 21$.
- $\text{Var}(X) = E(X^2) - \mu^2$ with discrete weights $f_i/150$.
- From \$28 upward, accumulate partial class [20,30).
- $\text{Var}(aX + b) = a^2\text{Var}(X)$.

Solution 5.8

(i) The total frequency gives

$$30 + a + 45 + b + 15 = 150 \quad \Rightarrow \quad a + b = 60.$$

The mean condition gives $\sum f_i x_i = 150(21) = 3150$. The known contributions are

$$30(5) + 45(25) + 15(45) = 1950,$$

so

$$15a + 35b = 3150 - 1950 = 1200.$$

Solving the simultaneous equations gives $\boxed{a = 45,\ b = 15}$.

(ii) Using the class centres,

$$E(X^2) = \frac{\sum f_i x_i^2}{150} = 585.$$

Since $\mu = 21$,

$$\mathrm{Var}(X) = E(X^2) - \mu^2 = 585 - 21^2 = \boxed{144}.$$

For the median, the cumulative frequencies are $30, 75, 120, 135, 150$. The middle position lies at the boundary between the second and third classes, so the median is estimated as $\boxed{\$20}$.

(iii) Within the $20 \le x < 30$ class, the portion above 28 is the interval from 28 to 30, which is $\frac{2}{10}$ of the class width. Assuming uniformity, this contributes

$$\frac{2}{10} \cdot 45 = 9$$

attendees. All attendees in the 30–40 and 40–50 classes also count, giving $9+15+15 = 39$. Hence

$$P(X > 28) \approx \frac{39}{150} = \boxed{0.26}.$$

(iv) $E(P) = 0.6(21) - 5 = 7.60$; $\mathrm{SD}(P) = |0.6|\sqrt{144} = 7.20$.

Takeaways 5.8

- Two equations recover missing table entries (mean + total count).
- Grouped variance uses centre-of-class plug-in expectations.
- Linear profit rules scale means and variances predictably.

5.3 Advanced

Problem 5.9: The exponential distribution and higher moments

Let X have the exponential density $f(x) = e^{-x}$ for $x \geq 0$ and $f(x) = 0$ otherwise. This distribution is often used to model waiting times. In this problem, you will compute its moments using a reduction formula rather than evaluating each integral from scratch.

(i) For $I_n = \int_0^\infty x^n e^{-x}\,dx$, prove $I_n = nI_{n-1}$ for $n \geq 1$.

(ii) Deduce $E(X^n)$, hence $E(X)$ and $\mathrm{Var}(X)$.

(iii) Skewness measures asymmetry and is defined by $E[(X-\mu)^3]/\sigma^3$. Compute its exact value here.

(iv) For $Y = \sqrt{X}$, show $E(Y) = \frac{\sqrt{\pi}}{2}$.

Hint:

- Integrate by parts with $u = x^n$, $dv = e^{-x}dx$.
- Moments equal factorials once $I_0 = 1$ is grounded.
- Expand cubic about μ.
- Substitute $u = \sqrt{x}$ and finish with Gaussian integrals; an integration by parts step gives $\int_0^\infty u^2 e^{-u^2}\,du = \frac{\sqrt{\pi}}{4}$.

Solution 5.9

(i) Use integration by parts with $u = x^n$ and $dv = e^{-x}\,dx$. Then $du = nx^{n-1}\,dx$ and $v = -e^{-x}$, so

$$I_n = \big[-x^n e^{-x}\big]_0^\infty + n\int_0^\infty x^{n-1}e^{-x}\,dx.$$

The boundary term is 0: at $x = 0$ it is zero for $n \geq 1$, and as $x \to \infty$ the exponential decay dominates x^n. Hence $I_n = nI_{n-1}$.

(ii) Since $I_0 = \int_0^\infty e^{-x}\,dx = 1$, repeated use of the reduction formula gives

$$I_n = n(n-1)\cdots 1\, I_0 = n!.$$

Therefore $E(X^n) = n!$. In particular,

$$E(X) = 1!, \qquad E(X^2) = 2!,$$

so

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 2 - 1 = 1.$$

(iii) Here $\mu = 1$ and $\sigma = 1$. Expand the central cubic:

$$E[(X-1)^3] = E(X^3) - 3E(X^2) + 3E(X) - 1.$$

Using $E(X^3) = 3! = 6$, $E(X^2) = 2$, and $E(X) = 1$,

$$E[(X-1)^3] = 6 - 3(2) + 3(1) - 1 = 2.$$

Dividing by $\sigma^3 = 1$ gives skewness $\boxed{2}$.

(iv) Since $Y = \sqrt{X}$,

$$E(Y) = \int_0^\infty \sqrt{x}\, e^{-x}\,dx.$$

Let $u = \sqrt{x}$, so $x = u^2$ and $dx = 2u\,du$. Then

$$E(Y) = \int_0^\infty 2u^2 e^{-u^2}\,du.$$

Using the standard Gaussian integral result $\int_0^\infty u^2 e^{-u^2}\,du = \frac{\sqrt{\pi}}{4}$, we obtain

$$E(Y) = \boxed{\frac{\sqrt{\pi}}{2}}.$$

Takeaways 5.9

- Reduction formulae replace one-off integration tricks.
- Polynomial moments encode factorial growth for this law.
- Γ-function values link $\mathbb{E}[\sqrt{X}]$ with Gaussian tails.

Problem 5.10: Moments and kurtosis of the standard normal

Kurtosis is a numerical measure of how much probability sits in the tails of a distribution compared with its centre. It matters because two distributions can have the same mean and variance but very different chances of producing extreme values. The normal distribution has kurtosis 3, which is often used as a benchmark when comparing tail behaviour.
Let $Z \sim N(0,1)$ with density $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$. For each integer $n \geq 0$, define

$$I_n = \int_{-\infty}^{\infty} z^n \phi(z)\, dz = E(Z^n).$$

This problem shows how symmetry and integration by parts generate the key normal moments.

(i) Argue $I_n = 0$ for odd n.

(ii) For even $n \geq 2$, derive $I_n = (n-1)I_{n-2}$ by parts.

(iii) Compute $\mathrm{Var}(Z)$ and $E(Z^4)$.

(iv) Kurtosis $\dfrac{E[(Z-\mu)^4]}{\sigma^4}$ for $Z \sim N(0,1)$.

Hint:

- Odd integrand $\times$ even ϕ is odd.
- Choose $u = z^{n-1}$, $dv = z\phi(z)dz$ with $v = -\phi(z)$.
- Start from $I_0 = 1$, step to I_2, I_4.
- Here $\mu = 0$, $\sigma = 1$.

Solution 5.10

(i) The density $\phi(z)$ is an even function because $\phi(-z) = \phi(z)$. If n is odd, then z^n is odd, so $z^n\phi(z)$ is odd. The integral of an odd function over the symmetric interval $(-\infty, \infty)$ is 0, hence $I_n = 0$ for odd n.
(ii) For even $n \geq 2$, write

$$I_n = \int_{-\infty}^{\infty} z^{n-1}\, z\phi(z)\, dz.$$

Since $\phi'(z) = -z\phi(z)$, we have $z\phi(z)\, dz = -d\phi(z)$. Integration by parts gives

$$I_n = \big[- z^{n-1}\phi(z)\big]_{-\infty}^{\infty} + (n-1)\int_{-\infty}^{\infty} z^{n-2}\phi(z)\, dz.$$

The boundary term vanishes, so $I_n = (n-1)I_{n-2}$.
(iii) Since $I_0 = 1$, the recursion gives

$$I_2 = (2-1)I_0 = 1, \qquad I_4 = (4-1)I_2 = 3.$$

Also $E(Z) = 0$, so

$$\operatorname{Var}(Z) = E(Z^2) - [E(Z)]^2 = 1 - 0 = 1.$$

(iv) For the standard normal, $\mu = 0$ and $\sigma = 1$. Therefore

$$\frac{E[(Z-\mu)^4]}{\sigma^4} = E(Z^4) = \boxed{3}.$$

This value is the reference kurtosis for a normal distribution.

Takeaways 5.10

- Gaussian moment recursion mirrors calculus reduction schemes.
- Symmetry zeros out odd moments instantly.
- Kurtosis 3 is the benchmark "mesokurtic" normal reference.

Problem 5.11: Statistical analysis of heart rates

Pulse rates vary from person to person and are often modelled approximately by normal distributions. Let $M \sim N(71, 9^2)$ model male pulse rate and $F \sim N(76, 9.5^2)$ model female pulse rate, with independence assumed where two people are compared. Let Φ be the standard normal cdf.

(i) Bradycardia: pulse < 60. Which sex has the larger population proportion below 60? Compare z-scores only.

(ii) Tachycardia: pulse > 100. Express $P(M > 100)$ using Φ. If mean becomes $71 - c$ and sd σ_{new}, state the equation keeping the same upper-tail probability as before.

(iii) Why is $P(\bar{M} > 100) < P(M > 100)$ for sample mean $\bar{M}$ of $n > 1$ males?

(iv) With $D = F - M$, find $E(D)$, $\operatorname{Var}(D)$, and $P(F < M)$ using Φ.

Hint:

- Compare standardized positions of 60 under both models.
- Tail probability $= 1 - \Phi(z)$; unchanged tail $\Rightarrow$ same z-threshold.
- Means have variance scaled by $1/n$.
- $F < M \iff D < 0$.

Solution 5.11

(i) $z_M = \dfrac{60-71}{9} = -\dfrac{11}{9}$ and $z_F = \dfrac{60-76}{9.5} = -\dfrac{32}{19}$ with $z_M > z_F$ (less extreme left tail for males). Thus $\Phi(z_M) > \Phi(z_F)$, i.e. males carry a larger proportion below 60 bpm.
(ii) $P(M > 100) = 1 - \Phi\left(\frac{29}{9}\right)$; unchanged tachycardia rate requires $\dfrac{100-(71-c)}{\sigma_{\text{new}}} = \dfrac{29}{9}$ i.e. $\dfrac{29+c}{\sigma_{\text{new}}} = \dfrac{29}{9}$.
(iii) For a sample mean of n independent male pulse rates,

$$\bar{M} \sim N\left(71, \frac{81}{n}\right).$$

The centre is still 71, but the standard deviation is $9/\sqrt{n}$, which is smaller than 9 for $n > 1$. Since 100 is far above the mean, the narrower sampling distribution places less area in the upper tail beyond 100. Hence $P(\bar{M} > 100) < P(M > 100)$.
(iv) Define $D = F - M$. Then

$$E(D) = E(F) - E(M) = 76 - 71 = 5.$$

Because F and M are independent, variances add even though the variables are subtracted:

$$\operatorname{Var}(D) = 9.5^2 + 9^2 = 171.25.$$

The event that a randomly chosen female has a lower pulse rate than a randomly chosen male is $F < M$, or $D < 0$. Therefore

$$P(F < M) = P(D < 0) = \Phi\left(\frac{0-5}{\sqrt{171.25}}\right).$$

Takeaways 5.11

- z-scores rank extremity across different normal models.
- Averages have tighter sampling distributions than singleton draws.
- Differences add variances even when subtracting independent variables.

6 Part 2: Practice and Challenge Bank

Part 2 presents additional problems. Try each problem first, then check the hint if needed. Each worked solution is followed by a short takeaways box to help you consolidate the main idea.

6.1 Warm-Up Drills

Problem 6.1: Two spinners

A toy wheel is labelled with the integers $1, 2, \ldots, 10$. On each spin, every label is equally likely, and a trial records the number shown. This warm-up checks that you can write a simple discrete uniform distribution and then use it to predict a count over repeated trials.

(i) Write the probability mass function for one spin, and find the probability that the label is divisible by 3.

(ii) If the wheel is spun independently 30 times, what is the expected number of spins that show a multiple of 3?

Hint:

- List residues mod3 among $\{1, \ldots, 10\}$.
- Expected count = trials $\times P$(event).

Solution 6.1

(i) Let X be the label shown on one spin. Since all ten labels are equally likely,

$$P(X = k) = \frac{1}{10}, \qquad k = 1, 2, \ldots, 10.$$

The multiples of 3 in this set are $3, 6, 9$, so

$$P(\text{multiple of 3}) = \frac{3}{10}.$$

(ii) If Y is the number of multiples of 3 in 30 spins, then $Y \sim B(30, \frac{3}{10})$. Therefore

$$E(Y) = np = 30 \cdot \frac{3}{10} = \boxed{9}.$$

Takeaways 6.1

- A discrete uniform distribution gives equal probability to each listed outcome.
- Count favourable labels first, then divide by the total number of labels.
- For repeated independent trials, the expected count is np.

6.2 Stretch Problems

Problem 6.2: Dice probabilities and proportion thresholds

A binomial/normal exercise on counts and proportions.

Two fair dice are thrown repeatedly. In each throw, call the result a "success" if the sum of the two dice is at least 10. Let X count the number of successes in n independent throws, and let $\hat{p} = X/n$ be the sample proportion of successes. This problem moves from the exact binomial model to the normal approximation, so be careful about when a continuity correction is being used.

(i) Find the exact single-throw success probability p, then determine the minimum number of throws n required so that $P(X \geq 2) > 0.90$.

(ii) With $n = 180$, approximate the probability that the number of successes is exactly equal to its expected value. Use a normal approximation to the binomial distribution with a continuity correction. You may use $\Phi(0.1) - \Phi(-0.1) \approx 0.0796$.

(iii) An observer wants the sample proportion $\hat{p}$ to fall within 0.05 of the true probability p with approximately 95% probability. Ignoring continuity corrections, find the minimum suitable n. Use $\Phi^{-1}(0.975) \approx 1.96$.

(iv) Show algebraically that applying a continuity correction of ± 0.5 to the count X is equivalent to applying a correction of $\pm\frac{1}{2n}$ to the proportion $\hat{p}$.

Hint:

- $P(X \geq 2) = 1 - P(0) - P(1)$; numerical search over n.
- "Exactly mean" corresponds to interval $[\mu - \frac{1}{2}, \mu + \frac{1}{2}]$ for the approximating normal.
- Margin $1.96\sqrt{p(1-p)/n}$.
- Divide inequalities by n.

Solution 6.2

(i) There are 36 equally likely ordered outcomes. The sums at least 10 are $(4,6),(5,5),(6,4),(5,6),(6,5),(6,6)$, so

$$p = \frac{6}{36} = \frac{1}{6}, \qquad q = 1 - p = \frac{5}{6}.$$

For $X \sim B(n, \frac{1}{6})$,

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1).$$

Thus we need

$$\left(\frac{5}{6}\right)^n + n\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^{n-1} < 0.10.$$

Testing integer values gives $n = 21$ too small, while $n = 22$ works, so the minimum is $\boxed{22}$ throws.

(ii) When $n = 180$, the binomial mean and variance are

$$\mu = np = 30, \qquad \sigma^2 = npq = 180 \cdot \frac{1}{6} \cdot \frac{5}{6} = 25.$$

Exactly 30 successes is represented by the continuous interval $29.5 \leq Y \leq 30.5$, where $Y \sim N(30, 25)$. Hence

$$z_1 = \frac{29.5 - 30}{5} = -0.1, \qquad z_2 = \frac{30.5 - 30}{5} = 0.1,$$

so the required probability is approximately

$$P(-0.1 \leq Z \leq 0.1) = \Phi(0.1) - \Phi(-0.1) \approx \boxed{0.0796}.$$

(iii) The approximate standard deviation of $\hat{p}$ is

$$\sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{5}{36n}}.$$

For a central 95% interval, require

$$1.96\sqrt{\frac{5}{36n}} \leq 0.05.$$

Squaring and rearranging gives $n \geq 213.42\ldots$, so the minimum integer is $\boxed{214}$.

(iv) Since $\hat{p} = X/n$ and $n > 0$, dividing the count inequality by n preserves the inequality directions:

$$x - \frac{1}{2} \leq X \leq x + \frac{1}{2} \iff \frac{x}{n} - \frac{1}{2n} \leq \frac{X}{n} \leq \frac{x}{n} + \frac{1}{2n}.$$

Therefore

$$\frac{x}{n} - \frac{1}{2n} \leq \hat{p} \leq \frac{x}{n} + \frac{1}{2n}.$$

Takeaways 6.2

- A binomial model begins with one clear success probability.
- Continuity correction turns a single count into a width-one interval.
- On the proportion scale, the correction shrinks from ± 0.5 to $\pm\frac{1}{2n}$.

Problem 6.3: Simulating a binomial by coin tosses (enrichment)

Enrichment: uses short Python; not examinable as a fixed-item in HSC.
Simulate $B(10, \frac{1}{2})$ as the count of heads in ten fair coin flips. One simulated experiment consists of ten flips; the recorded value is the number of heads. By repeating that experiment many times, you can see the empirical histogram cluster around the theoretical mean $np = 5$.

```
import random
N = 20000            # number of simulated experiments
results = []
for _ in range(N):
    heads = sum(random.random() < 0.5 for _ in range(10))
    results.append(heads)

# Optional: print mean and rough histogram
from collections import Counter
c = Counter(results)
print("mean count", sum(results)/N)
print("counts", dict(sorted(c.items())))
```

Task. Run the script, for example in Google Colab. Report the sample mean and comment on how close it is to $np = 5$. Explain why increasing N, the number of simulated experiments, sharpens the agreement between simulation and theory.

Hint:

- Each experiment is independent; mean of counts estimates np.
- Law of Large Numbers reduces Monte Carlo noise.

Solution 6.3

Each simulated experiment has expected count

$$E(X) = np = 10 \cdot \frac{1}{2} = 5.$$

Your printed sample mean will not usually equal 5 exactly, because the simulation itself is random. However, it should be close to 5 when $N = 20000$. If N is increased, the average is taken over more independent experiments, so the random simulation error tends to shrink. This is a computational version of the Law of Large Numbers.

Takeaways 6.3

- A binomial random variable can be simulated by repeating Bernoulli trials and counting successes.
- Simulation output is random, so exact agreement is not expected.
- Larger Monte Carlo sample sizes give more stable estimates.

Problem 6.4: Calculating normal probabilities with Python (enrichment)

Enrichment laboratory; software skills are illustrative.
Assume IQ scores in a population are modelled by $X \sim N(100, 15^2)$. The value 115 is one standard deviation above the mean. Use Python to compute the left-tail probability $P(X \leq 115)$, and compare the result with the standard normal table value $\Phi(1)$.

```
from scipy.stats import norm
print(norm.cdf(115, loc=100, scale=15))
```

Compare the output to $\Phi(1) = 0.8413$.

Hint:

- SciPy `norm.cdf` applies the cdf with explicit mean/scale arguments.
- $z = (115 - 100)/15 = 1$.

Solution 6.4

Standardising gives

$$z = \frac{115 - 100}{15} = 1.$$

Therefore

$$P(X \leq 115) = P(Z \leq 1) = \Phi(1) \approx 0.8413.$$

The Python command `norm.cdf(115, loc=100, scale=15)` applies the same calculation directly using the original mean and standard deviation, so it should print approximately 0.8413.

Takeaways 6.4

- Software cdf commands and standard normal tables are evaluating the same probability.
- The `loc` parameter is the mean; the `scale` parameter is the standard deviation.
- Standardising first is still useful for checking whether the output is reasonable.

Problem 6.5: Rare blue variants and sample proportions

In a breeding experiment, the true long-run proportion of blue offspring is believed to be $p = 0.20$. Let $\hat{p}$ denote the observed sample proportion. Because the sample sizes in this problem are large, we will approximate the distribution of $\hat{p}$ using a normal distribution.
(i) For $n = 500$, approximate the probability that strictly more than 22% of the flowers are blue. Use a continuity correction by translating "strictly more than 110 blue flowers" to the boundary 110.5.
(ii) Find the smallest integer n for which the normal approximation gives

$$P(|\hat{p} - p| < 0.03) \geq 0.99.$$

Hint:

- Carry counts to proportion scale before evaluating z on proportions.
- For (ii), set $z_{0.995} \approx 2.576$ (or cite your table) $\times \sqrt{pq/n}$.

Solution 6.5

(i) For the sample proportion,

$$\mu_{\hat{p}} = p = 0.20, \qquad \sigma_{\hat{p}} = \sqrt{\frac{pq}{n}} = \sqrt{\frac{0.2 \cdot 0.8}{500}}.$$

Strictly more than 22% means more than 110 blue flowers out of 500, so the continuity-corrected boundary is

$$\hat{p} > \frac{110.5}{500} = 0.221.$$

Thus

$$z = \frac{0.221 - 0.20}{\sqrt{0.16/500}} \approx 1.17.$$

The probability is approximately

$$P(Z > 1.17) = 1 - \Phi(1.17) \approx \boxed{0.12}.$$

(ii) A central 99% normal interval uses $z_{0.995} \approx 2.576$. We need the margin of error to be at most 0.03:

$$2.576\sqrt{\frac{0.2 \cdot 0.8}{n}} \le 0.03.$$

Rearranging,

$$n \ge \left(\frac{2.576\sqrt{0.16}}{0.03}\right)^2 \approx 1179.9.$$

Since n must be an integer, the minimum is $\boxed{1180}$ seeds.

Takeaways 6.5

- For sample proportions, use mean p and standard deviation $\sqrt{pq/n}$.
- A count-based continuity correction can be converted to a proportion boundary.
- High confidence and a small margin of error require a large sample size.

Problem 6.6: Advanced inheritance scenarios

A horticulturalist is attempting to cross two species of flowers to isolate a rare "true crimson" colour variant. According to the genetic model, each offspring independently has probability 0.25 of displaying the true crimson colour. For a large planting of 400 offspring, let $C \sim B(400, 0.25)$ count the number of crimson flowers. In part (i), use a smaller greenhouse count $X \sim B(20, 0.25)$.

(i) In the greenhouse of 20 offspring, find the conditional probability that exactly 5 flowers are true crimson, given that at least one flower is true crimson.

(ii) Determine the minimum number of offspring n that must be cultivated to ensure at least a 99% probability of obtaining two or more true crimson flowers.

(iii) For the commercial planting of 400 offspring, approximate $P(90 \leq C \leq 110)$ using a normal approximation with continuity correction.

(iv) A competing scientist tests a soil additive on 400 plants and observes 120 true crimson flowers. Find the uncorrected z-score relative to the baseline $p = 0.25$, and briefly comment on the evidence.

Hint:

- Conditional formula; complement for tail $1 - P(0) - P(1)$.
- Normal parameters $\mu = np$, $\sigma = \sqrt{npq}$.
- Two-sided continuity $(89.5,\ 110.5)$.

Solution 6.6

(i) Let $X \sim B(20, 0.25)$ for the greenhouse. Since the event $X = 5$ automatically implies $X \geq 1$,

$$P(X = 5 \mid X \geq 1) = \frac{P(X = 5)}{P(X \geq 1)}.$$

Now

$$P(X = 5) = \binom{20}{5}(0.25)^5(0.75)^{15} \approx 0.2023,$$

and

$$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.75^{20} \approx 0.9968.$$

Therefore

$$P(X = 5 \mid X \geq 1) \approx \frac{0.2023}{0.9968} \approx \boxed{0.2030}.$$

(ii) Let $C_n \sim B(n, 0.25)$. The complement of "two or more" is "zero or one", so

$$P(C_n \geq 2) = 1 - P(C_n = 0) - P(C_n = 1).$$

We need

$$0.75^n + n(0.25)(0.75)^{n-1} \leq 0.01.$$

Factoring gives

$$0.75^{n-1}(0.75 + 0.25n) \leq 0.01.$$

Testing nearby integers, $n = 23$ gives approximately 0.0116, which is still too large. For $n = 24$, the value is approximately 0.0090, so the minimum is $\boxed{24}$ offspring.

(iii) For $C \sim B(400, 0.25)$,

$$\mu = np = 100, \qquad \sigma = \sqrt{npq} = \sqrt{400(0.25)(0.75)} = \sqrt{75}.$$

The inclusive event $90 \leq C \leq 110$ becomes

$$89.5 \leq Y \leq 110.5, \qquad Y \sim N(100, 75).$$

The z-values are

$$z_1 = \frac{89.5 - 100}{\sqrt{75}} \approx -1.21, \qquad z_2 = \frac{110.5 - 100}{\sqrt{75}} \approx 1.21.$$

Hence

$$P(90 \leq C \leq 110) \approx P(-1.21 \leq Z \leq 1.21) \approx \boxed{0.774}.$$

(iv) Under the baseline model, 120 crimson flowers is compared with mean 100 and standard deviation $\sqrt{75}$. The uncorrected score is

$$z = \frac{120 - 100}{\sqrt{75}} \approx \boxed{2.31}.$$

This is more than two standard deviations above the baseline mean, so it would be considered unusually high under the 25% model and gives strong evidence that the additive may have increased the crimson rate.

Takeaways 6.6

- Conditioning removes part of the sample space; here it removes only the zero-crimson case.
- Minimum sample-size questions often use the complement and then test integers.
- Continuity corrections are essential when a normal curve approximates inclusive binomial counts.

Problem 6.7: Memoryless geometric distribution

Suppose independent trials are repeated until the first success occurs. Let X be the waiting time, counted as the trial number of the first success. If the success probability on each trial is p and $q = 1 - p$, then

$$P(X = x) = pq^{x-1}, \qquad x = 1, 2, \ldots$$

Assume $E[X] = 1/p$.

(i) Show $P(X > n) = q^n$.

(ii) Prove $P(X > n + k \mid X > n) = P(X > k)$, and briefly explain why this is called the memoryless property.

(iii) By differentiating the geometric series twice, show $\mathrm{Var}(X) = q/p^2$.

(iv) A space probe attempts to ping a satellite every 4 seconds. Each ping succeeds with probability 0.05. Find the mean and standard deviation of the total waiting time $T = 4X$ seconds.

Hint:

- Intersection $X > n+k$ inside $X > n$ is just the former event.
- Differentiate $\sum q^x$ or use known factorial moment route.
- $\mathrm{Var}(4X) = 16\mathrm{Var}(X)$.

Solution 6.7

(i) The event $X > n$ means the first n trials all fail. Since the trials are independent, this has probability q^n. Equivalently,

$$P(X > n) = \sum_{x=n+1}^{\infty} pq^{x-1} = pq^n(1 + q + q^2 + \cdots) = \frac{pq^n}{1-q} = q^n.$$

(ii) Using conditional probability,

$$P(X > n + k \mid X > n) = \frac{P(X > n + k)}{P(X > n)} = \frac{q^{n+k}}{q^n} = q^k.$$

But $q^k = P(X > k)$. This says that, after n failures, the probability of needing more than k additional trials is the same as it was at the start.
(iii) From

$$\sum_{x=0}^{\infty} q^x = \frac{1}{1-q},$$

differentiating twice gives

$$\sum_{x=2}^{\infty} x(x-1)q^{x-2} = \frac{2}{(1-q)^3}.$$

Multiplying by pq gives

$$E[X(X-1)] = \sum_{x=1}^{\infty} x(x-1)pq^{x-1} = \frac{2q}{p^2}.$$

Then

$$E[X^2] = E[X(X-1)] + E[X] = \frac{2q}{p^2} + \frac{1}{p},$$

so

$$\mathrm{Var}(X) = E[X^2] - [E(X)]^2 = \frac{q}{p^2}.$$

(iv) Here $p = 0.05$ and $q = 0.95$. Thus

$$E(X) = \frac{1}{0.05} = 20, \qquad \mathrm{Var}(X) = \frac{0.95}{0.05^2} = 380.$$

Since $T = 4X$,

$$E(T) = 4E(X) = 80 \text{ s}, \qquad \mathrm{SD}(T) = 4\sqrt{380} \approx \boxed{78 \text{ s}}.$$

Takeaways 6.7

- A geometric tail can be read as "all previous trials failed."
- Memorylessness means past failures do not change the future waiting-time distribution.
- Scaling time by 4 scales the standard deviation by 4 as well.

6.3 Challenge Corner

Problem 6.8: The mathematics of Benford's law

Many real-world data sets, such as populations, lengths, financial amounts, and physical constants, have leading digits that are not uniformly distributed. Benford's law models the leading digit $D \in \{1, \ldots, 9\}$ by

$$P(D = d) = \log_{10}\left(1 + \frac{1}{d}\right).$$

In this problem you will verify that this formula really defines a probability distribution and then connect it to a simple continuous model.

(i) Show that $\sum_{d=1}^{9} P(D = d) = 1$, so the formula is a valid probability mass function.

(ii) Show that the probability of a leading digit 1, 2, or 3 is greater than 60%.

(iii) By expanding $\sum dP(D = d)$ and regrouping logarithmic terms, prove that the mean leading digit is $\mu = 9 - \log_{10}(9!)$.

(iv) If $U \sim \mathrm{Unif}[0, 1)$ and $Y = 10^U$, show that the leading digit of Y follows Benford's law.

Hint:

- $\log(1 + \frac{1}{d}) = \log(d + 1) - \log d$.
- Sum three explicit logs to compare to $\log 4$.
- Expand expectations and regroup coefficients of $\log k$.
- Leading digit d iff $d \le Y < d + 1$; take $\log_{10}$.

Solution 6.8

(i) Rewrite each probability as

$$\log_{10}\left(1+\frac{1}{d}\right) = \log_{10}\left(\frac{d+1}{d}\right) = \log_{10}(d+1) - \log_{10} d.$$

Therefore the sum telescopes:

$$\sum_{d=1}^{9} P(D=d) = \log_{10} 2 - \log_{10} 1 + \cdots + \log_{10} 10 - \log_{10} 9 = \log_{10} 10 - \log_{10} 1 = 1.$$

(ii) The same telescoping idea gives

$$P(D \in \{1,2,3\}) = \log_{10} 2 + \log_{10}\frac{3}{2} + \log_{10}\frac{4}{3} = \log_{10} 4 \approx 0.602.$$

Thus the first three digits account for slightly more than 60% of all leading digits under Benford's law.
(iii) The mean is

$$E(D) = \sum_{d=1}^{9} d \log_{10}\left(\frac{d+1}{d}\right) = \sum_{d=1}^{9} d\big(\log_{10}(d+1) - \log_{10} d\big).$$

Expanding shows that most logarithms appear twice with neighbouring coefficients:

$$E(D) = 1\log_{10} 2 + 2\log_{10} 3 - 2\log_{10} 2 + \cdots + 9\log_{10} 10 - 9\log_{10} 9.$$

After regrouping, this becomes

$$E(D) = 9\log_{10} 10 - \log_{10}(1 \cdot 2 \cdots 9) = 9 - \log_{10}(9!).$$

(iv) The leading digit of Y is d precisely when

$$d \le Y < d+1.$$

Since $Y = 10^U$, taking $\log_{10}$ gives

$$\log_{10} d \le U < \log_{10}(d+1).$$

Because U is uniform on $[0,1)$, the probability is the length of this interval:

$$P(D=d) = \log_{10}(d+1) - \log_{10} d = \log_{10}\left(1+\frac{1}{d}\right).$$

Takeaways 6.8

- Logarithm differences often telescope into compact probability sums.
- Benford leading digits are heavily weighted toward smaller digits.
- Uniformity on a logarithmic scale produces Benford's law on the original scale.

Problem 6.9: The squared Cauchy distribution

Let

$$g(x) = \frac{1}{(1+x^2)^2}, \qquad -\infty < x < \infty.$$

This density-like curve has heavier tails than a normal distribution but still has finite mean and variance. The aim is to normalise it into a probability density function and then compute its first two moments carefully.

(i) Show $\displaystyle\int_{-\infty}^{\infty} g(x)\,dx = \frac{\pi}{2}$ using the substitution $x = \tan\theta$.

(ii) Verify that $f(x) = \dfrac{2}{\pi(1+x^2)^2}$ is a valid probability density function on the real line.

(iii) Show $E(X) = 0$, and justify why the improper integral converges before using symmetry.

(iv) Show that the variance is defined and that $\mathrm{Var}(X) = 1$.

Hint:

- Half-angle formulas after angular substitution.
- Non-negativity and total mass $(2/\pi)\cdot\pi/2 = 1$.
- Split $[0,\infty)$, substitute $u = 1 + x^2$ for the positive half before symmetrising.
- $E[X^2]$ via same substitution yields 1, then subtract 0^2.

Solution 6.9

(i) Let $x = \tan\theta$. Then $dx = \sec^2\theta\, d\theta$ and $1 + x^2 = \sec^2\theta$. As x runs from $-\infty$ to ∞, θ runs from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$. Hence

$$\int_{-\infty}^{\infty} \frac{1}{(1+x^2)^2}\, dx = \int_{-\pi/2}^{\pi/2} \frac{\sec^2\theta}{\sec^4\theta}\, d\theta = \int_{-\pi/2}^{\pi/2} \cos^2\theta\, d\theta = \frac{\pi}{2}.$$

(ii) The function is non-negative for all real x. Also,

$$\int_{-\infty}^{\infty} f(x)\, dx = \frac{2}{\pi}\int_{-\infty}^{\infty} \frac{1}{(1+x^2)^2}\, dx = \frac{2}{\pi}\cdot\frac{\pi}{2} = 1.$$

So f is a valid pdf.
(iii) The integrand for the mean is

$$xf(x) = \frac{2x}{\pi(1+x^2)^2},$$

which is odd. Before using symmetry, check convergence on the positive half-line:

$$\int_0^{\infty} \frac{2x}{\pi(1+x^2)^2}\, dx = \frac{1}{\pi}\int_1^{\infty} u^{-2}\, du = \frac{1}{\pi},$$

where $u = 1 + x^2$. The corresponding negative-half integral is finite and equal to $-\frac{1}{\pi}$, so $E(X) = 0$.
(iv) Since $E(X) = 0$, the variance is $E(X^2)$. Compute

$$E(X^2) = \int_{-\infty}^{\infty} \frac{2x^2}{\pi(1+x^2)^2}\, dx.$$

Using $x = \tan\theta$ again,

$$\frac{2}{\pi}\int_{-\pi/2}^{\pi/2} \tan^2\theta\cos^2\theta\, d\theta = \frac{2}{\pi}\int_{-\pi/2}^{\pi/2} \sin^2\theta\, d\theta = 1.$$

Therefore $\mathrm{Var}(X) = 1 - 0^2 = \boxed{1}$.

Takeaways 6.9

- A normalising constant is chosen so the total area under a pdf is exactly 1.
- Symmetry is powerful, but improper integrals still need convergence checks.
- The substitution $x = \tan\theta$ is natural whenever $1 + x^2$ appears.

7 Conclusion

Probability distributions are at the heart of statistics and data science. The techniques covered in this booklet — from reading a normal distribution table to applying the binomial model — appear regularly in HSC examinations and in real-world applications such as quality control, risk analysis, and machine learning.

Keep practising with a variety of problems, and remember: understanding the shape and parameters of a distribution is often the key to unlocking the solution.

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A The standard normal distribution

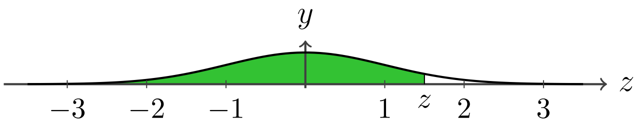
A brief summary of the standard normal probability distribution

The figure below shows the *standard normal probability density function* $y = \phi(z)$. The shaded region to the left of a vertical boundary at z corresponds to the *cumulative distribution function*

$$\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \phi(t)\, dt.$$

The table further down gives some values of $\Phi(z) = P(Z \le z)$. For example,

$$P(Z \le 1.6) = \Phi(1.6) = \int_{-\infty}^{1.6} \phi(z)\, dz \approx 0.9452.$$



z	**first decimal place** .0	.1	.2	.3	.4	.5	.6	.7	.8	.9
0.	0.5000	0.5398	0.5793	0.6179	0.6554	0.6915	0.7257	0.7580	0.7881	0.8159
1.	0.8413	0.8643	0.8849	0.9032	0.9192	0.9332	0.9452	0.9554	0.9641	0.9713
2.	0.9772	0.9821	0.9861	0.9893	0.9918	0.9938	0.9953	0.9965	0.9974	0.9981
3.	0.9987	0.9990	0.9993	0.9995	0.9997	0.9998	0.9998	0.9999	0.9999	1.0000

A more detailed table giving the values of z to three decimal places may be found in the appendix to Chapter 17 of many textbooks.

For many calculations the *empirical rule* (68–95–99.7) is sufficient:

$$P(-1 \le Z \le 1) \approx 68\%$$
$$P(-2 \le Z \le 2) \approx 95\%$$
$$P(-3 \le Z \le 3) \approx 99.7\%$$

